



CVR COLLEGE OF ENGINEERING

Department of CSE

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AI-Based Malware Detection System

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Introduction / Problem Statement

- Problem Domain: Cybersecurity & AI
- Problem Definition:
 - Traditional antivirus systems rely on signature-based detection.
 - They fail to detect zero-day and polymorphic malware.
- Objective: Develop an AI-powered malware detection system
 - using GANs, VAEs, and LLMs.

Recent Malware Attack Losses (Real-World Companies)		
Company / Organisation	Type of Attack	Reported Loss / Impact
CDK Global (2024)	Ransomware	\$25M ransom + ~\$605M dealer losses
Jaguar Land Rover (2025)	Ransomware	£196M direct + £1.9B broader impact
Allianz Life (2025)	Data Breach	1.1M customer data exposed
Mercer & Beacon Pointe (2026)	Data Theft	Legal & remediation costs
MGM Resorts (2023)	Ransomware	~\$100M revenue impact
Capital One (2023)	Payment Card Data Breach	\$25M settlement + \$1.8B in fines

Company / Organisation

Type of Attack

Reported Loss / Impact

CDK Global (2024)

Ransomware

\$25M ransom + ~\$605M dealer losses

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Allianz Life (2025)

Data Breach

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Mercer & Beacon Pointe (2026)

Data Theft

Existing Work / Challenges

- Existing Methods:
 - Signature-based detection
 - Heuristic-based detection
 - Rule-based IDS systems
- Challenges:
 - Cannot detect zero-day attacks
 - High false positive rate
 - Slow manual analysis
 - Malware evasion techniques

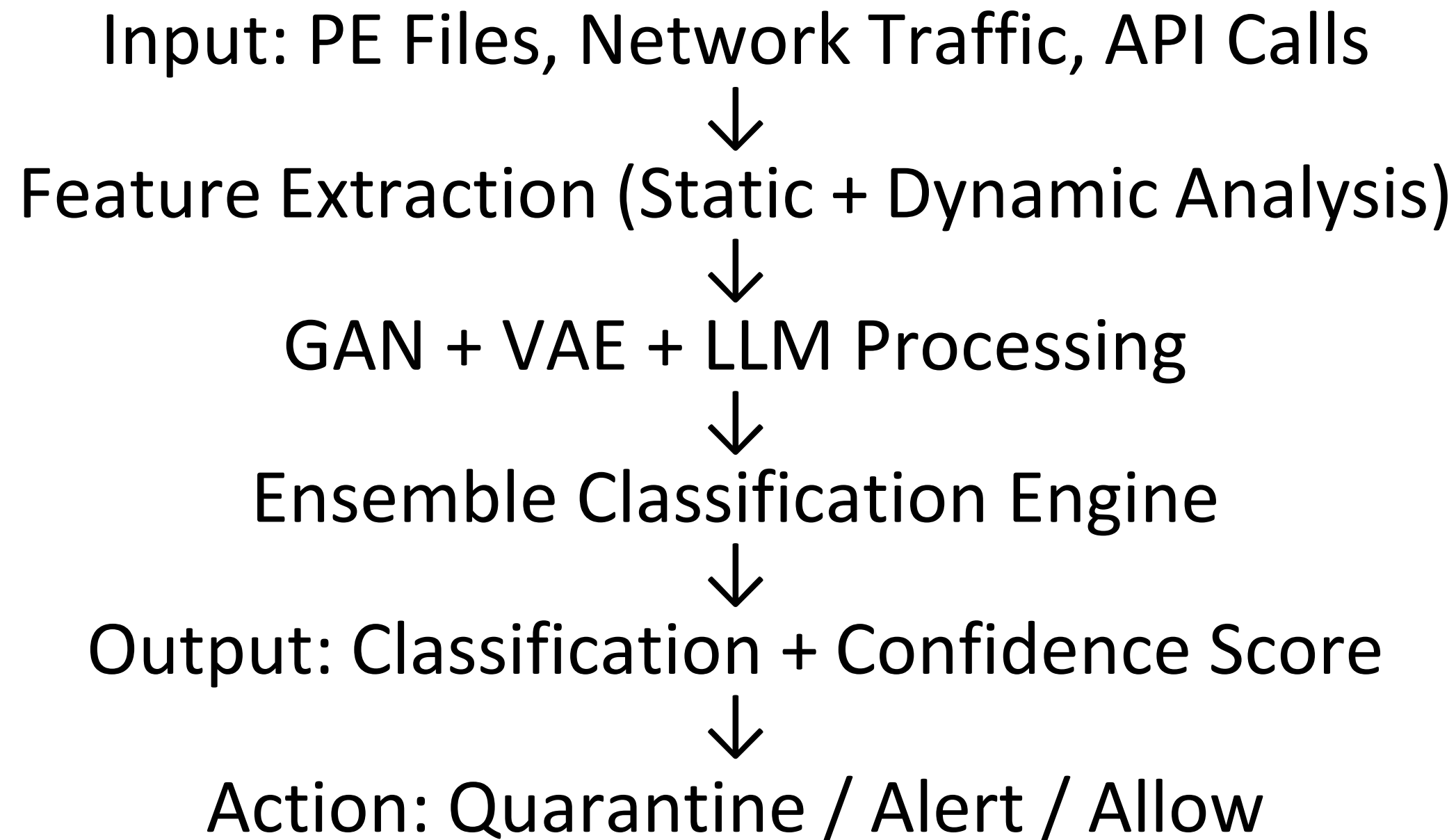
Literature Review / Related Works

- Deep learning improves malware detection accuracy.
- GAN-based models generate adversarial malware samples.
- VAE models detect anomalies using reconstruction error.
- LLMs assist in analyzing malicious scripts.
- Ensemble learning enhances detection performance.
- Recent research shows 94–97% detection accuracy using AI.

Proposed Method & Modules

- Proposed Method: Generative AI-Based Hybrid Detection System
 - Modules:
 - Data Collection Module
 - Feature Extraction Module
 - GAN Detection Module
 - VAE Anomaly Module
 - LLM Analysis Module
 - Response & Alert System

System Architecture Overview



Functional & Non-functional Requirements

- **Functional**

- Detect known & unknown malware
- Perform static & dynamic analysis
- Generate confidence score
- Trigger automated response
- Continuous model retraining
- Provide real-time alerts

- **Non-functional**

- High Accuracy ($\geq 97\%$)
- Low False Positive Rate ($\leq 2\%$)
- Real-time Processing ($< 200\text{ms}$)
- Scalability for enterprise deployment
- Secure data handling
- System reliability & availability

Hardware & Software Requirements

- **Hardware**

- Minimum 16GB RAM
- Intel i7 / Ryzen 7 Processor
- GPU (NVIDIA CUDA-enabled) for training
- 500GB+ storage for datasets
- Secure server for deployment

- **Software**

- Python 3.x
- TensorFlow / PyTorch
- Scikit-learn
- Pandas & NumPy
- Flask / FastAPI for deployment
- Windows / Linux OS